

理論経済学と社会の課題

Theoretical Economics and Social Challenges

経済学類講師・高梨誠之

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教員紹介/Self-introduction

- Completed the doctoral programme and received a Ph.D in 2017.
- In 2022, I joined the Faculty of Economics and Management, Institute of Human and Social Sciences, Kanazawa University.
- During my undergraduate years, I was interested in mathematics, and later became interested in game theory.
- From the master's course, I studied economics.
- I research game theory and decision theory.
 - Basically, I am researching theoretical models in economics.

経済学について/About economics

- In the first place, what is economics?
- "The study of how **economic agents** choose to **allocate (scarce) resources** and the impact of these **choices** on society" (アセモグル、レイブソン、リスト(2020)) (translated by DeepL)
- An economic agent is an individual or group that makes choices.
 1. **A consumer** chooses to buy a hamburger or a cheeseburger.
 2. **A parent** chooses whether to send their child to a public or private high school.
 3. **A citizen** chooses whether or not to vote.
 4. **A company** chooses whether or not to donate to a political party.
 5. **A political party** chooses whether or not to submit a bill to parliament.

経済学について/About economics

- If resources are plentiful, there is no need to consider how to distribute them.
 - We just need to give everyone what they want.
- The current situation is not one where there are abundant resources.
 - It would take 2.8 Earths to provide everyone in the world with the same standard of living as people in Japan today. (Global Footprint Network (2018))
- There is a need for discussion on how to allocate limited resources (scarce resources).

経済学について/About economics

- The distribution of resources is not decided by a single economic agent.
- Let's consider the market for electric vehicles.
- The car company decides how many electric cars to produce.
- Consumers decide whether or not to purchase an electric vehicle from among gasoline-engine cars, hybrid cars, etc.
- The government chooses which types of vehicles to subsidise.
- The allocation of resources is (in most cases) determined by the interaction of multiple economic agents.

経済学について/About economics

- The allocation of resources is determined by the choices of multiple economic agents.
- The study of economics is the study of what kind of choices economic agents make, what kind of allocation is determined by those choices, and what kind of impact such choices and allocations have on society.
 - Within economics, there are various fields that focus more on choice, more on distribution, more on the impact on society, etc.

経済理論がなぜ必要か

Why is an economic theory necessary?

- All of the things I study are theoretical models of the economy.
- In short, the theoretical model is a "virtual world" created mathematically.
 - An image of a virtual reality space created for people's economic activities. (?)
 - (Although in reality it is much simpler).
- There is absolutely no guarantee that this virtual world and the real world will coincide.
 - Most theoretical economists, myself included, do not believe that they will coincide.
 - In fact, there are people who are economically successful without even knowing about theory.
 - There is a lot of criticism of economic theory.
- So, do we not need theoretical models?

経済理論がなぜ必要か

Why is an economic theory necessary?

- The world's economic activities are extremely complex.
- As a result, the economic problems that need to be solved are extremely complex and diverse.
 - To begin with, nobody (may) knows how to solve them.
- Facing each of those problems empty-handed is impossible for anyone except a "genius."
 - If an amateur tries to solve the problem, they may make it worse or more complicated.
- A theoretical model that serves as a standard or benchmark can be a "weapon".
 - It can provide clues to solutions and perspectives that have been overlooked.
 - Even if it doesn't solve the problem, it may be possible to prevent it from getting worse, or even make it a little better.

経済理論モデルの例

Example of an economic theory model

- What models are there for the world's economic models?
- Here, we will introduce a model of a strategic-form game from game theory.
- Let N be the set of players, S_i the set of strategies for each player $i \in N$, and the payoff function $u_i: \prod_{j=1}^{|N|} S_j \rightarrow \mathbb{R}$.
- The model of a strategic game is written as $(N, \{S_i\}_{i \in N}, \{u_i\}_{i \in N})$.
 - In short, the model of a game is described by (the set of) **players**, (the set of) each player's **strategy**, and (the function of) each player's **payoff**.

モデルによる分析方法

Analysis Method Using Models

- I will give an example of analysing real economic phenomena using models.
- We will introduce an auction called the English Auction.
 - Bidders compete to raise the price, and when no new bids are made, the bidder with the highest bid at that point wins the auction.
- In order to make analysis more clearly, let's consider the following.
 - All bidders raise their hands at the starting price of 0 yen.
 - The price rises continuously from 0, and as long as the bidders are willing to buy, they continue to raise their hands, and when they run out, they lower them.
 - The auction stops the moment there is only one person left holding up their hand, and the remaining person becomes the winner, who pays the price at the time of the stop.

モデルによる分析方法

Analysis Method Using Models

- Each participant thinks about when to lower their hand.
 - In other words, the price at which a participant lower your hand represents her/his **strategy**.
- Let's assume there are three participants in the auction.
 - Let's call them **Participants 1, 2 and 3**.
- Let's assume that participants 1, 2 and 3 each raise their hands at 1 million yen, 1.3 million yen and 0.8 million yen respectively.
 - This price is the amount each participant is willing to pay.
- When the price is 1 million yen, the only person who has their hand up is Participant 2.
- Participants 1 and 3 receive no **payoff**, while participant 2 receives 300,000 yen.

モデルによる分析方法

Analysis Method Using Models

- Generally, in this auction, the auction stops at the second highest willingness-to-pay (WTP) among all participants.
 - The highest bidder wins and pays the second highest WTP.
- This is true only for this auction; different results may be obtained under the rules of other auctions.
 - This means that the outcome would be different if the rules of another game were used.
- In this way, many economic phenomena have been explained by analysing each individual game (not just auctions).
- A tool that only explains reality?

社会課題の解決—メカニズムデザイン—

Solving Social Challenges -Mechanism Design-

- In recent years, there has been a lot of attention paid to a field called mechanism design (or market design).
- This is a field that does not simply create and analyse real-life models, but analyses what kind of rules would produce better results.
- If I had to compare it to something, it's a field that creates VR that everyone can enjoy, rather than creating VR that resembles reality.(?)

社会課題の解決—メカニズムデザイン—

Solving Social Challenges -Mechanism Design-

- Let's explore some examples of the application of mechanism design.
 - Based on Prendergast (2017).
- Let's take a look at the story of the American non-profit organisation "Feeding America".
- The aim of Feeding America is to distribute donated food "appropriately" to local "food banks".
 - Food banks are charitable organisations that distribute food to people in poverty who do not have enough to eat.
- Feeding America is an organisation that oversees more than 200 food banks.
 - Feeding America distributes food to each food bank.

社会課題の解決—メカニズムデザイン—

Solving Social Challenges -Mechanism Design-

- Before 2005, the system used was a "waiting list" system.
- Problems with the "waiting list" system.
 1. Despite the fact that each food bank has different needs in terms of the amount and type of food required, this is not being addressed.
 - Food that is valuable to one food bank is distributed elsewhere.
 - For example, milk is sent to a place where the fridge is already full.
 2. It is not possible to check which food banks are prepared to accept food at the same time.
 - It takes time to confirm the need for and ability (such as whether they can transport it) of the food bank that has contacted you.
 - In the meantime, the food may go bad.

社会課題の解決—メカニズムデザイン— Solving Social Challenges -Mechanism Design-

- Since 2005, a choice system that applies auction theory has been adopted.
 - Not only did it solve the problems of the waiting list system, but it also sought to create a system that would prevent small food banks from being exploited.
- A unique currency called "Share" was created, and distributed to each food bank according to a rule, and then the food was auctioned off.
 - This was because using actual currency (dollars) would create disparities in fundraising ability.
 - There is no other use for the “shares” other than this auction.
 - It is also possible for each food bank to auction off food.

社会課題の解決—メカニズムデザイン—

Solving Social Challenges -Mechanism Design-

- Each truckload is auctioned twice a day after the contents of the food have been displayed.
- Bidding at the auction is done once only, and bidders cannot see the bids of other bidders.
- The person who bids the highest amount wins the auction and pays the amount.
- It is also possible to bid jointly with two food banks.
 - The food from one truck is divided into two lots.
- Food banks with surplus food can also auction off the food.

社会課題の解決—メカニズムデザイン—

Solving Social Challenges -Mechanism Design-

- They found that the price of each food item varied greatly, indicating that there are significant differences in demand.
 - The most expensive cereal is about 50 times more expensive than the cheapest produce.
- The choice system has made it easier to allocate high-priced and low-priced products to the food banks that need them.
- By allowing food banks to auction off food, the supply increased by about 23,000 tons in one year.
- The scarce resource of food was more appropriately **allocated** based on the **choices** made by **each food bank**.

社会課題の解決—メカニズムデザイン—

Solving Social Challenges -Mechanism Design-

- It is likely that in some cases, food banks do not actually act in accordance with theory when placing bids.
- Also, when designing the system, it was not created by simply applying a theory, but consideration was given to real-life situations.
- However, the latter was superior to the former, which was a "waiting list" system created without the theory, and a choice system created with the theory.
- Perhaps this is how the benefits of theory are demonstrated.

まとめ/Summary

- We introduced some of the research methods used in theoretical economics.
- We introduced the kind of research theoretical economics is doing to solve social challenges.

参考文献/References

1. アセモグル、レイブソン、リスト (2020) 「アセモグル/レイブソン/リスト ミクロ経済学」 東洋経済新報社
2. Global Footprint Network (2018) “NFA2018 Edition” (<https://data.world/footprint/nfa-2018-edition/>, retrieved April 24, 2022)
3. Prendergast, Canice, “How food banks use markets to feed the poor,” *Journal of Economic Perspectives*, 2017, 31 (4), 145–62.